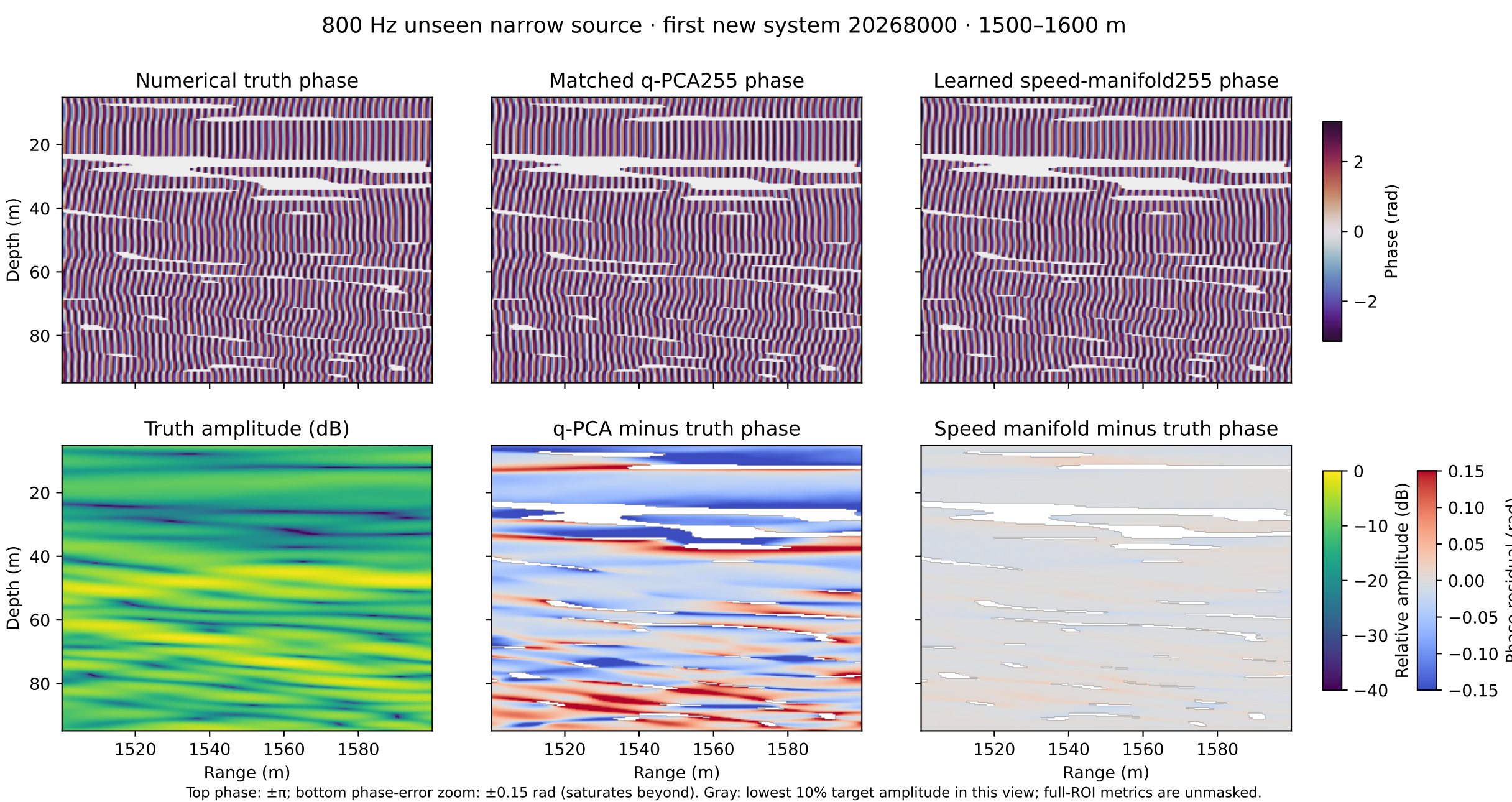

800 Hz unseen narrow source · first new system 20268000 · 1500–1600 m
Numerical truth phase
Matched q-PCA255 phase
Learned speed-manifold255 phase
Truth amplitude (dB)
q-PCA minus truth phase
Speed manifold minus truth phase
Depth (m)
20
40
60
80
1520
1540
1560
1580
Range (m)
Phase (rad)
2
0
−2
Relative amplitude (dB)
0
−10
−20
−30
−40
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Top phase: ±π; bottom phase-error zoom: ±0.15 rad (saturates beyond). Gray: lowest 10% target amplitude in this view; full-ROI metrics are unmasked.